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B.TECH
(SEM VI) THEORY EXAMINATION 2022-23
ANALOG SIGNAL PROCESSING

Time: 3 Hours ***Total Marks: 100***

Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. 2 x 10 = 20

(a) Briefly describe the active filter and its uses.
(b) Draw the circuit of high pass filter using op-Amp.
(c) Find the value of Chebyshev co-efficient $C_2(\omega)$.
(d) Write only two properties for the given Buttreworth response

$$|T_n(j\omega)|^2 = \frac{1}{1+\omega^{2n}}$$

(e) What is meant by constant-resistance lattice?
(f) Which filter is used for delay equalization and how they minimize it.
(g) Define FDNR and its application.
(h) Define the LC ladders circuit and why its energy dissipation is zero?
(i) Draw the grounded and floating resistor with the use of OTA
(j) Draw the circuit diagram of summer using op- Amp.

SECTION B

2. Attempt any *three* of the following: 10x3=30

(a) Design a three op-amp based biquad filter. Also, drive all the standard filter responses.
(b) Explain the magnitude responses of the Buttreworth and Chebyshev with suitable example.
(c) Explain the procedure of equalization of first order filter.
(d) Briefly explain the Bruton's FDNR technique with neat sketch.
(e) Design the switched capacitor cascaded filter with suitable example.

SECTION C

3. Attempt any *one* part of the following: 10x1=10

(a) Explain the current conveyor and its advantages over the op-amp. Also, design the integrator and differentiator circuits using current conveyor.
(b) Describe the Sallen-key biquad filter. Also, derive the filter responses

4. Attempt any *one* part of the following: 10x1=10

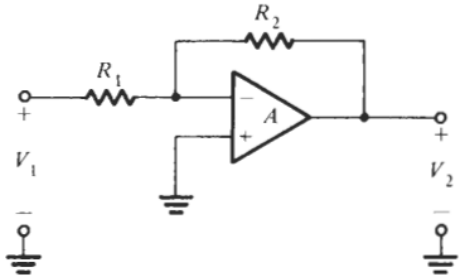
(a) Describe the Butterworth filter and design the first order Butterworth low pass filter using Op-amp with the derivation of its transfer function.
(b) Comparison of Maximally flat and Equal ripple responses.

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QP23EP1_311 | 21-06-2023 08:54:53 | 139.167.199.53

5. Attempt any *one* part of the following: 10x1=10

(a) Find out the transfer function of the inverting amplifier (considering op-amp gain=A)and then calculate the sesitivity of the inverting amplifier w.r.t R_1 (assume A=∞)



(b) Explain the strategies for Equalizer design

6. Attempt any *one* part of the following: 10x1=10

(a) Explain the Gorski-Popiel's Embedding Technique with suitable example.

(b) Draw the circuit of a Generalized Impedance Convertor (GIC). Realize a grounded impedance using GIC and find its value.

7. Attempt any *one* part of the following: 10x1=10

(a) How is the higher order filters formed? From the given specifications, determine the voltage gain of second order high pass butter worth filter
$R_3=R_2=33\Omega$
$f=250H_z$
$f_L=1KH_z$

(b) Write a short notes on:
(i) Amplifiers
(ii) Gyrator. Also, draw the circuits using OTA device.

QP23EP1_311 | 21-06-2023 08:54:53 | 139.167.199.53